

27 An electron has a kinetic energy of 1.0 MeV. If its momentum is measured with an uncertainty of 1.0%, what is the uncertainty in its position?

- A** $7.7 \times 10^{-10} \text{ m}$ **B** $1.2 \times 10^{-10} \text{ m}$ **C** $2.9 \times 10^{-12} \text{ m}$ **D** $4.1 \times 10^{-19} \text{ m}$

Ans: (B)

$$E = \frac{p^2}{2m}$$

$$p = \sqrt{2mE} = \sqrt{2(9.11 \times 10^{-31})(1.0 \times 10^6 \times 1.6 \times 10^{-19})}$$
$$= 5.399 \times 10^{-22} \text{ kg m s}^{-1}$$

$$\Delta x \cdot \Delta p \gtrsim h$$

$$\Delta x \gtrsim \frac{h}{\Delta p} = \frac{6.63 \times 10^{-34}}{0.01 \times 5.399 \times 10^{-22}} = 1.2 \times 10^{-10} \text{ m}$$

28 When the number of protons and the number of neutrons in a nuclide are both “magic numbers”